I am going to do my first python program

shift + enter to run (Shortcut)

```
print("Hello World")
```

```
Hello World
```

Double-click (or enter) to edit

What is Numpy?

Numerical Operation Library in python

```
# !pip install numpy
```

```
import numpy as np
```

```
votes = np.array([ 775,  787,  918,   88,  166,  286, 2556,  324,  504,  402])
costs = np.array(["800.0" ,"800.0", "800.0", "300.0", "600.0", "600.0", "600.0", "700.0" ,"550.0", "500.0"])
```

```
print("Votes (Array): ",votes )
```

```
Votes (Array):  [ 775  787  918   88  166  286 2556  324  504  402]
```

```
print("Costs (Array):", costs)
```

```
Costs (Array): ['800.0' '800.0' '800.0' '300.0' '600.0' '600.0' '600.0' '700.0' '550.0'
 '500.0']
```

```
type(votes)
```

```
numpy.ndarray
```

```
type(costs)
```

```
numpy.ndarray
```

```
# How is this diffrent from Python List?

a = [1,2,3,4]

type(a)
```

```
list
```

```
b = np.array([1,2,3,4])

type(b)
```

```
numpy.ndarray
```

```
## Python list can hold hetrogenous values (It can hold differnt datatypes)

e = ["Aryan", 1, 1.34, True]

e
```

```
['Aryan', 1, 1.34, True]
```

```
## Numpy array only allows Homogenous Values (Same datatypes)

w = np.array(["Aryan", 1, 1.34, True])

w
```

```
array(['Aryan', '1', '1.34', 'True'], dtype='<U32')
```

```
np.array([1, 1.34, True])
```

```
array([1.  , 1.34, 1.  ])
```

```
## It goes by a data types priority: String > Float > Integer > Boolean
```

```
np.array([1, 1.34, "Aryan", True])
```

```
array(['1', '1.34', 'Aryan', 'True'], dtype='<U32')
```

```
## Python is genral programming language.


# Python is slow (True)

# Python is easy to learn.
```

```
a = [1,2,3]
b = [1,2,3,4]

print(a,b)
```

```
[1, 2, 3] [1, 2, 3, 4]
```

```
a = 23
b = 34

a
```

```
23
```

```
person = ["Aniruddha", 28, "Data Engineer", True]
# [name, age, role, is_active]
```

Dimension and Shape

```
type(np.array([1, 1.34, True]))
```

```
numpy.ndarray
```

```
votes
```

```
array([ 775,  787,  918,   88,  166,  286, 2556,  324,  504,  402])
```

```
## How to know the dimension?
votes.ndim
```

```
1
```

```
## How to know the shape of an array?

votes.shape
```

```
(10,)
```

```
## How to calculate the number of elements in my array?
votes.size
```

```
10
```

```
## Create a 2d array.

r = np.array([[1,2,3],
              [4,5,6]])
```

```
r.ndim
```

```
2
```

```
r.shape
```

```
(2, 3)
```

```
r.size
```

```
6
```

3D array looks like?

```
# fruit image
!gdown 17tYTDPBU5hpby9t0kGd7w_-zBsbY7sEd
```

```
Downloading...
From: https://drive.google.com/uc?id=17tYTDPBU5hpby9t0kGd7w_-zBsbY7sEd
To: /content/fruits.png
100% 4.71M/4.71M [00:00<00:00, 24.5MB/s]
```

```
import matplotlib.pyplot as plt

img = np.array(plt.imread('fruits.png'))
```

```
img.ndim # R G B
```

```
3
```

```
img.shape
```

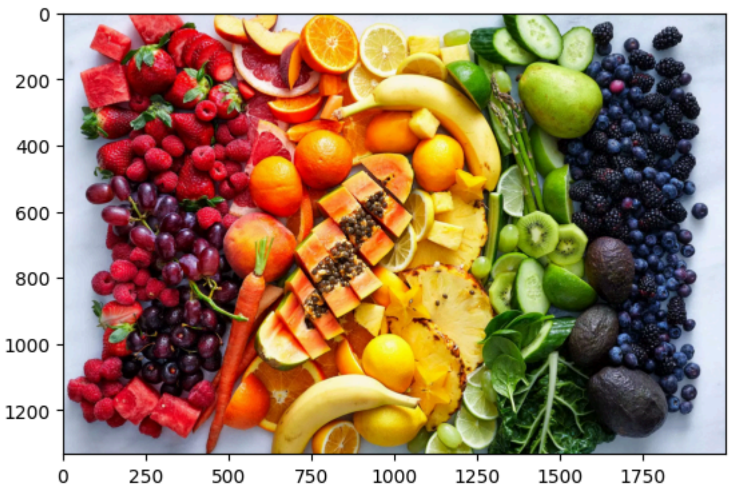
```
(1333, 2000, 3)
```

```
type(img)
```

```
numpy.ndarray
```

```
plt.imshow(img)
```

```
<matplotlib.image.AxesImage at 0x789df2d10500>
```




```
img
```

```
array([[[0.8784314 , 0.9137255 , 0.972549  ],
        [0.8784314 , 0.9137255 , 0.972549  ],
        [0.8784314 , 0.9137255 , 0.972549  ],
        ...,
        [0.8       , 0.85490197, 0.9098039 ],
        [0.8       , 0.85490197, 0.9098039 ],
        [0.8       , 0.85490197, 0.9098039 ]],

       [[0.8784314 , 0.9137255 , 0.972549  ],
        [0.8784314 , 0.9137255 , 0.972549  ],
        [0.8784314 , 0.9137255 , 0.972549  ],
        ...,
        [0.8       , 0.85490197, 0.9098039 ],
        [0.8       , 0.85490197, 0.9098039 ],
        [0.8       , 0.85490197, 0.9098039 ]],

       [[0.8784314 , 0.9137255 , 0.972549  ],
        [0.8784314 , 0.9137255 , 0.972549  ],
        [0.8784314 , 0.9137255 , 0.972549  ],
        ...,
        [0.8039216 , 0.85882354, 0.9137255 ],
        [0.8039216 , 0.85882354, 0.9137255 ],
        [0.8039216 , 0.85882354, 0.9137255 ]],

       ...,

       [[0.74509805, 0.79607844, 0.87058824],
        [0.74509805, 0.79607844, 0.87058824],
```

```
         [0.74509805, 0.79607844, 0.87058824],
         ...,
         [0.83137256, 0.8627451 , 0.9411765 ],
         [0.83137256, 0.8627451 , 0.9411765 ],
         [0.83137256, 0.8627451 , 0.9411765 ]],

        [[0.74509805, 0.79607844, 0.87058824],
         [0.74509805, 0.79607844, 0.87058824],
         [0.74509805, 0.79607844, 0.87058824],
         ...,
         [0.83137256, 0.8627451 , 0.9411765 ],
         [0.83137256, 0.8627451 , 0.9411765 ],
         [0.83137256, 0.8627451 , 0.9411765 ]],

        [[0.74509805, 0.79607844, 0.87058824],
         [0.74509805, 0.79607844, 0.87058824],
         [0.74509805, 0.79607844, 0.87058824],
         ...,
         [0.83137256, 0.8627451 , 0.9411765 ],
         [0.83137256, 0.8627451 , 0.9411765 ],
         [0.83137256, 0.8627451 , 0.9411765 ]]], dtype=float32)
```

Type Conversion

```
a = np.array(['1.2', '2.5', '3.6', '4.8'])
print("Type of a: ", type(a))
```

```
Type of a:  <class 'numpy.ndarray'>
```

```
print("Type of elements of a: ", a.dtype)
```

```
Type of elements of a:  <U3
```

```
b = a.astype(float)
```

```
type(b)
```

```
numpy.ndarray
```

```
b.dtype
```

```
dtype('float64')
```

Quiz

```
arr = np.array([1.4, 2.8, 3.9])
```

```
arr.astype(int)
```

```
array([1, 2, 3])
```

```
arr.astype(bool)
```

```
array([ True,  True,  True])
```

```
arr.astype(str)
```

```
array(['1.4', '2.8', '3.9'], dtype='<U32')
```

Doubts

```
a = [1,2,4,5,6]

b = np.array(a)

b
```

```
array([1, 2, 4, 5, 6])
```

```
person = ["string", int, float, bool]
```

```
# variables

# functions
```

```
# Class

# variables -> properties/class variables
# functions --> methods
```

```
# np.sum()

# votes.size
```

Tuple is a datastructure in python

Array is the datastructure of numpy

Tuples cannot be modified after creation, arrays can be modified after creatation

```
y = np.array(["abc", "10", "20"])

y.astype(int)
```

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
/tmp/ipython-input-1246339937.py in <cell line: 0>()
      1 y = np.array(["abc", "10", "20"])
      2 
----> 3 y.astype(int)

ValueError: invalid literal for int() with base 10: np.str_('abc')
```

Next steps: Explain error

```
# image ---> array (CNN)

# table ---> arrary

# text ---> array (NLP)



# post lecture attachment.
```

```
int32
```

```
int
```

```
Linked List --> Node (add, data)
```

```
w = [1,2,3,4]

for items in w:
  print(items)
```

```
1
2
3
4
```

```
Start coding or generate with AI.
```